Flood Modelling and Stormwater Design Guideline

Revision 1

Stormwater & Waterways Management

The Hills Shire Council

February 2024

REVISION CONTROL

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Foreword

This guideline applies to all developments proposed within The Hills Shire Council (Council) LGA. The document's main purpose is to outline Council's requirements for the design of stormwater drainage and the conduct of flood impact assessments (FIA) in support of development applications and planning proposals.

It is not intended to be a comprehensive or a stand-alone design manual. Rather, it should be read in conjunction with relevant policies, codes, industry guidelines, and standards, including those listed under Section 1 of this document. This document takes precedence over other Council design guidelines related to stormwater design and FIA. Where discrepancies exist between this guideline and other Council documents, the applicant is advised to contact Council's Development Assessment team to determine the guideline applicable to the development proposal.

Council reserves the right to regularly update this document to ensure its completeness, currency and adherence to industry best practice. Applicants and consultants are encouraged to refer to this guideline prior to undertaking any modelling or design work. Compliance with this guideline and the lodgement of a complete submission will assist Council in expediting the assessment of a development application or planning proposal.

Applications submitted without the required information will be returned or rejected. By providing adequate and correct information at the time of lodgement, the applicant will avoid the need for the Council to request additional information. The submission of all required information at lodgement can ensure that your application is processed in a timely manner.

All information, statement, or advice expressed or implied in this publication is provided in good faith. Council makes no representation or warranty of any kind, express or implied, regarding the accuracy, adequacy, validity, reliability, availability or completeness of any information provided in this document.

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1. Relevant Guidelines and Standards

The design shall be carried out with reference to and in accordance with:

- THSC Design Guidelines for Subdivision and Developments, 2011;
- The Institution of Engineers, Australia, Australian Rainfall and Runoff, A Guide to Flood Estimation, 1987;
- The Commonwealth of Australia and the Institution of Engineers, Australia, Australian Rainfall and Runoff, A Guide to Flood Estimation, 2019;
- NSW Government's Flood Risk Management Manual – The Management of Flood Liable Land, April 2005, particularly NSW State Government's Flood Prone Land Policy;
- NSW Government's Floodplain Development Manual – The policy and manual for the management of flood liable land, 2023:
- Engineers Australia, Water Engineering, Australian Rainfall & Runoff Project 11 Report – Blockage of Hydraulic Structures - Blockage Guidelines (February 2015);
- Engineers Australia, Water Engineering, Australian Rainfall & Runoff Project 15 Report – Two Dimensional Modelling in Urban and Rural Floodplains;
- Floodplain Risk Management Guidelines by Office of Environment and Heritage (OEH, formally DECCW);
- NSW Department of Primary Industries - Office of Water (Controlled Activities under the Water Management Act 2000 on Waterfront Land) Guidelines for Watercourse Crossings on Waterfront Land;
- Landcom, Managing Urban Stormwater: Soils and Construction, Volume 1, 2004 (the Blue Book)
- The Hills Shire Council, Development Control Plan (DCP), 2012; Part C, Section 6, Flood Controlled Land;
- Australian Standard AS1597.1-2010 (Small) and AS1597.2-2013 (Large) – Precast reinforced concrete box culvert;
- Australian/New Zealand Standard AS/NZS 4058-2007 for precast concrete pipes; and
- The Department of Primary Industries preferred waterway crossing design documented in "Why do Fish Need to Cross the Road? Fish Passage Requirements for Waterway Crossing" (NSW Fisheries 2003).

2. Scope of Detailed Drainage Design

Scope of drainage investigation and design includes:

- Review of existing studies, reports, models and other relevant information;
- Hydrologic modelling to provide all required inputs for the drainage design;
- Catchment modelling for drainage systems;
- Modelling of water courses;
- Hydraulic design of cross drainage structures including culverts and bridges as required;

- Design of roadway drainage systems;
- Satisfactory management of surface runoff;
- Mapping to display flooding affectation and impacts on flood level and flow velocity;
- Preparation of design report; and
- Risk and hazard of flood impacts is minimised and made safe.

3. Design AEP

Drainage infrastructure shall be designed with the following design AEP flood event:

Item	AEP Flood
Stormwater systems (general)	10%
Stormwater systems (sag)	5%
Overland flow including road surface drainage systems	1%
Design of major cross drainage structures, including culverts and bridges	1%
Channel forming flows for bridges and culverts	39.35% to 20%
Open channels designed as major drainage system	1%
Flood evacuation route road and bridges	PMF

Design should be based on the fully developed catchment conditions.

4. Infrastructure Requirements

4.1 Road Surface Flow

For the design AEP events the following requirements have to be satisfied as a minimum:

- The maximum water depths at any point on the road must not be more than 4mm in the travel lanes, at intersections and on auxiliary lanes on the approaches to interchanges and intersections; and
- The maximum change in the depth of flow across the road must not exceed 4mm over 10 metres.

4.2 Surface Drainage

The design shall have sufficient provision for the drainage of all surface water, including the following specific requirements:

- Drainage must be provided where the works intercept runoff, floodplains, watercourses, depressions or drainage lines;
- Inlet and outlet treatments shall be designed in accordance with the Blue Book to avoid scour, undercut or erosion;

- Batter catch drains must be provided up-slope of all cut batters and at the foot of fill embankments where the ground slopes towards the embankment. Batter and catch drains are to be connected to the drainage system; and
- Open drains shall have suitable treatments to maintain stability and safeguard against erosion, determined by flow velocity and sheer stress.

4.3 Pits and Pipes

Drainage pipes and pits must comply with the following criteria and guidelines:

- Separate cross drainage systems from road drainage systems;
- The minimum drainage pipe sizes allowed are 300mm for longitudinal drainage and 450mm for cross drainage systems;
- The pipe and pit network should be designed to be self-cleansing. A minimum pipe grade of 1.0 % shall be provided for this purpose. A maximum grade of pipelines shall be in accordance with Table 4.12 of the Design Guidelines;
- Blockage factors of 0.5 and 0.2 are to be applied for sag pits and on-grade pits respectively;
- Check for any conflict between existing and proposed utilities with the proposed drainage system. In case of conflict being identified, the Consultant must coordinate with utilities and propose a satisfactory response;
- The depth of drainage pits must provide for connection of the subsoil drainage systems;
- 90-degree connections with the road drainage network are not permitted;
- Specify the required class of pipe and its installation type in accordance with the Concrete Pipe Association's "Concrete Pipe Selection and Installation" guide. The use of HS3 type installation is the minimum permitted;
- Consider soil conditions - systems should be designed to allow for settlement where soft soils are an issue;
- Pipes must be spigot and socket types with rubber ring seal;
- Include fall protection on headwalls and other structures where there is a risk of injury due to falling but ensure this doesn't introduce a collision hazard for vehicles; and
- Where there is a risk of debris blockage, pits should be designed with all-weather access to enable structures to be maintained and cleared.

4.4 Culverts

Culvert and bridge hydraulic design (major drainage systems) shall comply with the following requirements:

- Unless otherwise stated culvert/bridge openings are sized to convey up to 1% AEP from upstream catchments flood event. The soffit of the proposed structure must provide minimum 500mm freeboard above the 1% AEP flood level at the upstream side of the crossing;
- Allowance for blockage is to be in accordance with the requirements of the publication "Australian Rainfall & Runoff Revision Project 11: Blockage of Hydraulic Structures - Blockage Guidelines" (February 2015);

- The design must also ensure no increase in upstream/downstream flood levels by more than those levels determined or established for the existing catchment conditions as approved by Council;
- The design must also ensure no increase of velocities by constricting flows in developed catchment conditions; and
- All cross drainage structures must be provided with in-situ concrete headwalls, wingwalls and aprons at both inlet and outlet sides of the structure in accordance with THSC standard drawings SD.8 and SD.9.

Specific Design Requirements - Culverts

- Culvert base design and invert levels in Key Fish Habitat areas are to be "fish friendly". The Consultant shall liaise with NSW Fisheries to determine any Key Fish Habitat and necessary requirements;
- Base of the openings are not to create any additional flood or erosional impacts on upstream and downstream;
- Make provision for fauna passage in line with relevant legislative requirements; and
- Provide all weather access to the outlet for maintenance, where the hydraulic design of a culvert is outlet controlled and there is a risk of siltation.

4.5 Bridges

- Design to be substantially in accordance with the principles set out in the publications *"Australian Standard Bridge Design Part 1: Scope and General Principles"* (AS 5100.1-2004 (Incorporating Amendment No.1)), and *"AUSTROADS Guide to Bridge Technology Part 4: Design Procurement and Concept Design"*;
- Minimum freeboard to the underside of the bridge structure is to be in accordance with the requirements of the abovementioned publications, having regard for carriageway cross-falls;
- Bridge clear opening to be a minimum of 50% of the overall width of the riparian corridor, measured from toe of abutments, and allowing for bridge pier widths;
- Piers are to be designed to be streamlined in the direction of flow. Other than in unavoidable circumstances, no piers are to be constructed so as to obstruct the primary waterway area *(between low banks)*;
- Bridges are to be low energy style structures, minimising afflux at the design flood *(1% AEP, post-development case)*;
- Allowance for blockage is to be in accordance with the requirements of the publication *"Australian Rainfall & Runoff Revision Project 11: Blockage of Hydraulic Structures - Blockage Guidelines"* (February 2015);
- Consideration may be made for the construction of *"relief"* culverts through each abutment to account for the allowance of blockage in the bridge design;
- Hydraulic modelling is to be undertaken to determine the two dimensional (2-D) flow behaviour for channel forming flows *(consider the 39.35% AEP and 20% AEP floods)* and the design flood *(1% AEP)* to enable the design of any bed, bank and abutment scour protection works;
- Bridge pier scour depths are to be estimated and appropriate bed protection works are to be incorporated in the design;

- Flood levels predicted for the 1% AEP post development *(bridge included)* scenario are to be utilised in the designs for bulk earthworks including landfill adjoining any riparian corridor.

5. Floodplain and Waterways Requirements

5.1 Floodplain Compensatory Storage Requirements

Proposed fill related to new development that results in a loss of floodplain storage within the site up to 1% AEP Water Surface Elevation (WSE) must be compensated for with new and equivalent storage. In relation to Compensatory Storage, the applicant must provide evidence that the design meets the following minimum requirements.

- The total flood storage available within the site below the 1% AEP WSE under developed conditions must be equal to or greater than the storage available under existing conditions.
- The allocation of compensatory storage will not cause an increase in WSE within the site during flood events up to the 1% AEP flood. Additionally, the proposed development and allocation of compensatory storage must not increase WSE, flow velocities, and VxD in areas downstream of the site or elsewhere during flood events up to the 1% AEP flood. Limited exceptions will be allowed, particularly near major new structures.
- The loss of storage must be accommodated within the major drainage system of the subject site, i.e., hydraulically linked with the main flow path/s. Refer to Figure 1 below for details.

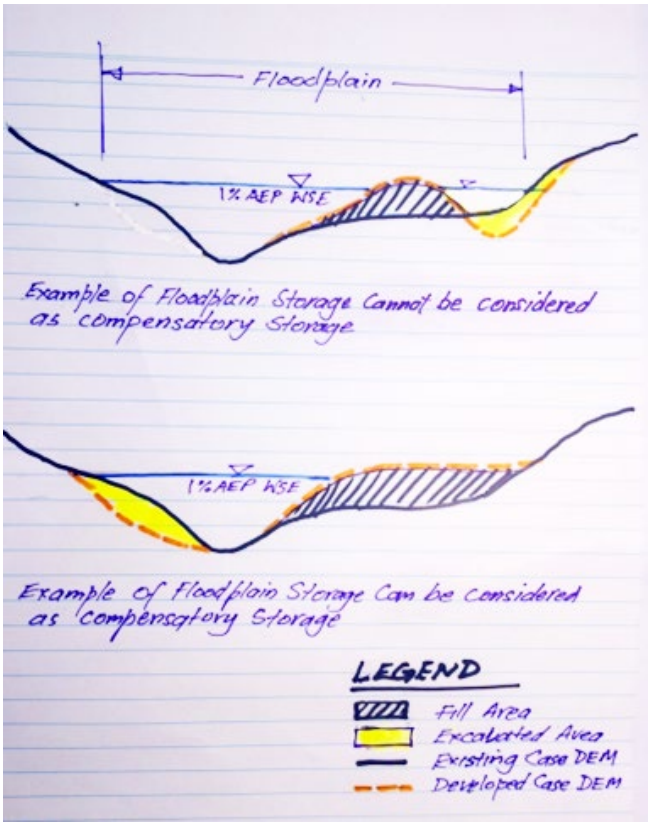


Figure 1. Requirements for compensatory floodplain storage.

5.2 Works Within a Waterway Corridor

- Stormwater outlets have to be aligned with channel flows;
- Channel works are to be designed in line with fluvial geomorphology principles and
- processes. It is recommended that a person qualified or experienced in fluvial geomorphology and riverine engineering is engaged in the design and construction of works within waterway corridors;
- Waterway corridors are to be designed to replicate natural processes, conditions and characteristics found local to the subject site. Constructed stormwater drainage systems are to consider this requirement and be designed in such a way to enable this;
- In addition to engineering and geomorphic requirements, waterway corridor works are to consider habitat and vegetation to achieve a fully functioning riparian ecosystem and incorporate these elements in engineering design plans; and
- Stream order is not to change as a result of upstream urbanisation and piping of watercourses.

5.3 DRAFT Waterways Safety Design Guidelines for Trunk Drainage (Open Channel) Systems

The safety of people can be compromised when exposed to flows which exceed their ability to remain standing and/or traverse a waterway. Australian Rainfall and Runoff (AR&R) set some guidelines/limits based on velocity and depth of overland flows to prevent pedestrians being swept through overland flow paths/open channels during major storm events (up to 1%AEP event).

The following table provides flow hazard regimes for infants, children and adults. These hazard regimes for tolerable flow conditions, i.e. the product of velocities and depths (DV) as related to the individual's physical characteristics (H.M).

Flow Hazard Regimes for People

<table>
<tr><th>DV (m2s-1)</th><th>Infants, small children (H.M ≤ 25) and frail/older persons</th><th>Children (H.M = 25 to 50)1</th><th>Adults (H.M > 50)</th></tr>
<tr><td>0</td><td>safe</td><td>safe</td><td>safe</td></tr>
<tr><td>0 – 0.4</td><td rowspan="5">Extreme Hazard; Dangerous to all</td><td>Low Hazard</td><td rowspan="2">Low Hazard</td></tr>
<tr><td>0.4 – 0.6</td><td>Significant Hazard; Dangerous to most</td></tr>
<tr><td>0.6 – 0.8</td><td rowspan="3">Extreme Hazard; Dangerous to all</td><td></td></tr>
<tr><td>0.8 -1.2</td><td>Significant Hazard; Dangerous to most</td></tr>
<tr><td>> 1.2</td><td>Extreme Hazard; Dangerous to all</td></tr>
</table>

Following Safety Design Criterion for trunk drainage/overland flow paths has been set by The Hills Shire Council (THSC) in accordance with the AR&R and The NSW Floodplain Development Manual.

- In a new trunk drainage/overland flow paths maximum Depth x Velocity (DV) product of 0.4m^2/s will be maintained where possible.
- Where DV is greater than 0.4m^2/s, the design will be required to specifically provide additional measures (as stated below) for the safety of persons who may enter, either voluntarily or by accident, into the channel/trunk drainage.
- Trunk drainage systems should be designed as much as possible to prevent persons falling or being swept into them, and to permit easy escape (provision of safe egress points) from flowing channels. Wide, grass lined channels are preferable to hard-lined channels with steep or vertical sides.
- Side slopes should be gradual, so persons will not be suddenly plunged into deep water. Slopes of 1:4 (vertical:horizontal) or flatter are recommended for grass-lined channels
- Rails or fences should be provided at transitions and other locations where steep slopes are necessary.
- For steep channels, fencing and warning signs will be required to prevent persons falling into the channel in wet or dry conditions.
- While it is usually impractical to fence channels completely, rails or fences must be provided at the most dangerous sections such as those near schools or upstream of culverts and closed conduits. These should not impede potential rescuers. Minimum fence height of 1.2 m should be used.
- Warning signs must be erected at strategic locations alerting people to the possible hazards. Signs should inform the public of the function of the drainage facility, as well as giving a warning for possible danger/hazards.
- Depth gauge boards also should be provided on channels, as an indicative hazard/warning as well as a means of recording information.
- Where the channel slope is too steep to achieve the desired safe hazard regimes, pools and riffles or drop structures can be built across the channel to achieve shorter and flatter reaches and hence to retard flow velocities.
- Drainage channels can be vegetated but needs to be regularly maintained. However, no large trees should be placed within the channel unless located in an area of low velocity.
- Provide a 1.5m wide access/safety strip along at least one side of the channel above the design flood level.

- Culverts at the downstream end of open channels should be provided with inlet screens.

5.4 Determination of Riparian Corridor (RC) Widths

Council recommends the following procedure to determine Riparian Corridor (RC) Widths.

Recommended Riparian Corridor (RC) Widths

Watercourse Type[1]	Vegetated Riparian Zone (VRZ)[2] Width	Total RC width
1st order	10 m	20 m+ bank full channel width[3]
2nd order	20 m	40 m+ bank full channel width
3rd order	30 m	60 m+ bank full channel width
4th order and greater (includes estuaries, wetlands and any parts of rivers influenced by tidal waters)	40 m	80 m+ bank full channel width

1 As classified under the Strahler System of ordering watercourses
2 Refer to Figure 2 for detail
3 Bank full channel width is often determined based on width of flood extent from a 39.35% AEP event

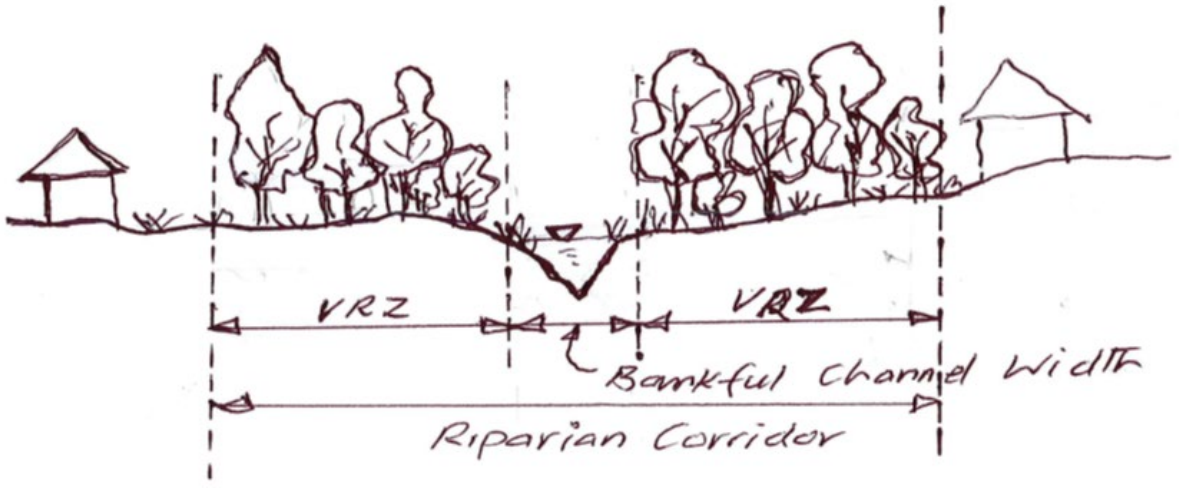


Figure 2. Riparian corridor definition.

The width of the VRZ should be measured from the bank on both sides of the watercourse.

[Additional Note: Channel width is not added in the Determination of Flood Control Lot Buffer, and the buffer width is measured from the centreline of the creek on each side of the creek.]

6. Hydrologic and Hydraulic Modelling Requirements

6.1 Corporate Software

The corporate hydrologic and hydraulic modelling programs used by Council are DRAINS, HEC-RAS(1D), TUFLOW, and RAFTS. The applicant/consultant is to utilise the latest versions of these programs to undertake any design works.

The sections below provide the minimum requirements and specific guidelines that need to be complied with by the applicant/consultant when using Council's corporate software for their modelling purposes.

6.2 DRAINS

6.2.1 Model Setup

Following are the minimum requirements for DRAINS Modelling:

- Use the latest version of DRAINS model software.
- DRAINS model is to be run in the standard hydraulic mode.
- Apply Council's blockage factors for inlet pits. Blockage factors of 0.5 and 0.2 are to be applied for sag pits and on-grade pits respectively.
- Pit inlet capacities should be derived based on pit lintel and grate openings.
- Downstream model boundary should be extended sufficiently downstream of the study area boundary so that backwater effects from the boundary condition have minimal impacts on the hydraulic grade line
- Apply water levels of the receiving waterbody as downstream boundary if known. Otherwise, the following general downstream boundary condition will apply:
    - Up to 20% AEP storm event, outlet pipe is to be free draining;
    - Greater than 20% AEP but less than 5% AEP storm event, use downstream obvert of outlet pipe as downstream condition; and
    - Greater than 5% AEP storm event, add a metre above the downstream obvert level of the outlet pipe.
- Pipe Roughness/Friction - A default Colebrook-White value of 0.6mm is to be adopted for old (existing) pipes. A Colebrook-White value of 0.3mm should be used when modelling new pipes during the concept design phase.
- Pit Pressure Loss Coefficients –The Queensland Urban Drainage Manual (QUDM) method, as automated in DRAINS, can be adopted to derive the pit pressure loss coefficients. However, the QUDM method is not always appropriate in the assessment of existing urban drainage systems. Should the QUDM method be adopted, sensitivity checks at locations where the networks are sensitive to head-loss must be undertaken.
- Where Ku values are greater than 4 in the DRAINS model, sensitivity checks are to be undertaken and the use of a higher Ku value justified to Council.
- QUDM methods are not to be utilised for the first pit in stormwater network line. Pit pressure loss coefficients for these pits are to refer to the results of the publication "Pressure Change at Storm Drain Junctions" (Smerdon et. al.), known as the "Missouri Charts".
- Council's preferred method of sensitivity assessment is to refer to the "Missouri Charts".

- Impervious Percentage – Use Table 4.6 of the Design Guidelines for Subdivision/Developments, 2011, The Hills Shire Council (available on-line at http://www.thehills.nsw.gov.au/Engineering-Specifications.html)
- Overland Flow Time (to peak) – In determining overland flow time Council recommends utilising either the "Friends" formula or the Kinematic Wave Equation with suitable surface resistance coefficients.
- Design Events – Run model for the 50%, 20%, 10% 5%, 2% and 1% AEP storms and the Probable Maximum Flood (PMF) events.
- As an initial order of magnitude check, the model results shall be reviewed using alternative methods such as the Probabilistic Rational Method (AR&R) and the Regional Flood Frequency Estimation (RFFE) method.
- Models shall be run for a range of storm durations sufficient to identify the critical storm.
- Modelling parameters recommended for use are as follows:
    - Impervious area depression storage = 1 mm
    - Supplementary area depression storage = 1 mm
    - Pervious area depression storage = 5 mm
    - Soil Type = 3
    - AMC = 3

6.2.2 Model Checks (Sensitivity)

Sensitivity analysis shall be carried out to assess how much influence the model parameter values have on the results. The principal parameters are rainfall losses, catchment storage and lag, friction, energy losses and pipe roughness. The sensitivity of the model results to downstream boundary conditions shall also be tested.

6.2.3 Naming Convention - DRAINS Modelling Files

DRAINS models developed and submitted as part of a DA must follow the following naming convention.

Segment	Description	Example
1	Catchment Number	C36
2	Catchment Name	Stevensons (not included in the name)
3	Subcatchment reference – street name or another identifier	Model B or Carole Ave
4	Event ID	50%, 20%, 10% 5%, 2% and 1% AEP storms and the Probable Maximum Flood (PMF) events
5	Duration	Including duration is optional and would be applicable if the model run for a single duration (i.e., critical duration storm) only

6	Scenario/Catchment Condition	EX- Existing IMP – Improved FC- Full Capacity
7	QA Reviewed	QA
8	Version Name	V01, V02, V03, etc.

Example file name: C36_CaroleAve_1%AEP_120min_EX_QA_V01.drn

6.2.4 DRAINS Modelling Quality Assurance Log

The applicant is to use a modelling log to maintain a record of model development, traceability, and quality assurance. An example of standard modelling is provided in Appendix E.

The modelling log is to be submitted with each draft modelling review and the final model.

6.3 HEC RAS (1D)

The following are areas in which 1D modelling using HEC RAS (1D) program can be accepted by Council. HEC-RAS (2D) is not a Council corporate software at this stage.

- Waterways and floodplains in which the flow is one-dimensional in nature, i.e., the dominant flow direction and forces follow the general waterway flow path;
- Steep streams that are highly gravity driven and have small overbank areas; and
- Areas in which the basic data does not support the potential gain of using a 2D model.

6.3.1 Model Setup

Following are the minimum requirements for HEC-RAS Modelling:

- The latest version of HEC-RAS should be used unless otherwise justified.
- Model geometry must be georeferenced to appropriate spatial coordinates.
- Model cross sections must be drawn at a regular interval and at locations where there is a significant topographical variation to appropriately represent ground conditions and waterway configuration.
- The point where the reach line intersects each cross section is to be set to station zero. Cross section transverse stations will be positive to the right of the reach line and negative to the left, looking downstream.
- Downstream model boundary of the study area must extend as far downstream as necessary to clearly establish the suitable boundary condition at the study area's downstream end.
- The study limits must extend a sufficient distance upstream of the study area to clearly demonstrate the impacts of the proposed development on the upstream properties.
- A description of and a rational explanation for how the upstream and downstream boundary conditions were established must be provided in the design report.
- A description of the Manning's n (channel roughness) values used and justification for why they were chosen must be provided.

- All proposed grading, culverts, bridges, drop structures, access ramps, etc. that impact on water surface profiles must be modelled.
- Buildings and other permanent obstructions within flowpaths are to be accurately represented in the model. Options of ineffective flow areas and blocked obstructions are to be used to represent or approximate the resulting effects of structures or constrictions.
- Energy loss through major culverts and bridges should be checked and confirmed using an alternate calculation, such as using a different tool, for example, HY-8, or by hand calculation.
- Run model simulations based on hydrological inputs for the 50%, 20%, 10%, 5%, and 1% AEP design storms are to be undertaken and reported graphically and in tables.
- HEC-RAS program has the provision to add descriptions within the model. Adequate descriptions should be added wherever possible to record information on data entry, versions, plans and amendments, etc.

6.3.2 Model Checks (sensitivity, check using the alternate program)

Sensitivity analysis shall be carried out to assess how much influence the model parameter values have on the results. The principal parameters are storage, channel roughness and energy losses, etc. Variation of parameters for this exercise to be limited to ±20%.

6.4 TUFLOW

6.4.1 Model Setup

The minimum requirements for TUFLOW modelling are as follows:

TUFLOW Version

Where possible the latest version of TUFLOW model is to be used.

Digital Elevation Model (DEM)

DEM for the study area to be developed using ALS data and/or site survey data.

Grid Size

Model grid size is to represent the flow behaviour in an urban environment, with consideration of narrow overland flowpaths, such as between buildings and permanent obstructions. One meter grid size is recommended unless a larger grid size is justified.

Roughness and Rainfall Losses

Council standard TUFLOW materials file (*.tmf*) to assign Manning's n and other parameters (such as rainfall losses) to land use areas is attached in Appendix D. Applicants to use this standard file in setting up TUFLOW models. Deviation from the standard parameters must be consulted with Council's Waterways Team and pre-approved before being utilised.

Flowpath Obstructions and Constrictions

Obstructions across a flowpath, such as buildings, sheds, fences, road embankments etc. are to be appropriately incorporated into the model with reference to the recent physical modelling

undertaken as part of *Australian Rainfall & Runoff - Revision Projects and Document Updating Project 15 - Two-Dimensional (2D) Modelling in Urban Areas*. Details on how to model buildings and fences are included in the following sections.

Downstream Boundary

Downstream boundary conditions in the TUFLOW model may be defined using one of the flowing approaches:

- Assigning a water level versus flow curve (HQ Curve); or
- Assigning a water level versus time curve (HT Curve).

The available hydraulic models from previous studies may be used to obtain these relationships for the drainage catchments.

The model boundary should be justified and extended sufficiently downstream of the study area boundary so that backwater effects from the boundary condition have minimal impacts on predicted flooding behavior.

Upstream Flow Boundary

Upstream flow boundary must be located away from the study area to eliminate the influence of the boundary on the modelling results.

For single lot developments hydrograph generated using the standard procedure at the downstream site boundary must be applied as upstream flow boundary.

Council may be able to provide hydrograph data, if available for the subject site. Standard fees and charges in line with Councils Fees and Charges Schedule will apply.

Initial Water Level (IWL)

Set up appropriate initial conditions for each event to be modelled. Pre-wetting of the catchment can be applied as constant initial water levels (IWL). Allocated IWLs to be commensurate with the starting water level of downstream water level boundaries.

Design Events

Minimum 1% AEP event to be run for both the existing and the developed catchment conditions. Simulation of more frequent events, such as 5%, 10%, and 20% AEP storms may also be required. Modelling of the PMF event will be required for critical and sensitive uses facilities and developments.

Please consult with Council's Waterways Team for further advice.

Cumulative Mass Error

- It is to be demonstrated that the TUFLOW model is fit-for-purpose through the assessment of the allowable mass balance error percentage value;
- All three Cumulative Mass Error percentage values such as for the overall model, for all the 2D domains and for any 1D domains should be within a ±3% limit;

- It is considered that some models will show a high mass error at the start of the simulation, which can be acceptable, provided it diminishes quickly; and
- The model will not be accepted by Council where the Cumulative Mass Error is outside the specified ±3% range.

Flood/Overland Flow Path Mapping

- Cut off depth of 0.1m to be used for mapping flood extents;
- At least 0.2m contour intervals to be used in flood level and flood depth mapping.

Flood/Overland Flow Path Mapping

- Cut off depth of 0.1m to be used for mapping flood extents.
- At least 0.2m contour intervals to be used in flood level and flood depth mapping

6.4.2 Provision of Model Files

TUFLOW input files and folders could be corrupted or missed while manually compiled and copied on a USB/Hard Drive. In the interest of copying the complete set of input and output files without any corruption it is recommended that the following procedure is followed:

- All the model files are copied using the in-built 'copy' function in TUFLOW, as described under Section 11.5.2.1 and in *Table 11-2 of the 2016 TUFLOW Manual*. This ensures that all input files that are necessary to run the model are copied into one folder;
- Any file read as a binary XF file corresponding original input data file needs to be manually copied into the folder;
- In addition to the input files, a complete set of output files are to be provided for all the scenarios/options run to ensure that the same results are produced when the files are rerun.
- The folder structure is to be in accordance with the *TUFLOW User Manual Build 2018, Section 2.1, Suggested Folder Structure.*

6.4.3 Modelling Buildings in TUFLOW

Buildings in floodplains may be represented by one of the following methods:

- Increasing bed roughness (high roughness)
- Blocking out of 2D elements or raising elements
- Modelling the building's exterior walls and leaving a break in the wall to allow water to enter (preferred by Council)
- Using energy loss coefficients to represent the fine-scale energy dissipation within and around the building. This is an alternative to using increased bed roughness
- Modelling buildings as "Porous" by partially blocking the 2D element sides within the building outline.

Methodology of modelling buildings needed to be discussed with Council prior to adopting in the final model.

Details on the preferred option are included in Appendix – B.

6.4.4 Modelling Fences/Walls in TUFLOW

Fences in floodplains may be modelled by one of the following methods:

- Modelling fences as a thin or zero width obstruction, i.e., does not affect storage
- 2d_zsh layers to model permanent fences (example: brick wall), using a thin width obstruction, i.e., does not affect storage.
- Variable z shape (2d_vzsh) layers to model fences that collapse under pressure (example: paling fence). Details on the methodology can be found in the TUFLOW manual.
- Variable z shape (2d_vzsh) layers to model fences that collapse under pressure (example: paling fence). Details on the methodology can be found in the TUFLOW manual.
- Proposed boundary fences along the overland flowpaths should be permeable. Fencing should not create any impediment to the flow of floodwaters.

Methodology of modelling buildings needed to be discussed with Council prior to adopting in the final model.

6.4.5 Sensitivity Analysis and Checks with Alternative Methods

Sensitivity analyses shall be carried out to assess how much influence the model parameter values, and key assumptions have on the results. The sensitivity analysis should include the impact assessment of the following parameters and assumptions:

- Rainfall losses used;
- Roughness and modelling buildings on the floodplain;
- 2D model grid size; and
- Downstream boundary condition.

The Applicant is to clearly state the proposed approach and methodology along with the nature and extent of sensitivity testing that will be undertaken.

The Applicant shall obtain Council's concurrence on appropriate model parameter values prior to proceeding with design runs.

[*Note (Future Consideration): Adding limits on variation, for example –*

- *Rainfall losses: ±20% variation*
- *Roughness: ±20% variation*

What about considering of effects of climate change on rainfall by increasing design rainfall by 20%, for example?]

6.4.6 Representation of PO Lines and PO Points

Following are the minimum requirements for representing PO Lines and PO Points in TUFLOW.

PO Lines

1. MUST be digitised from left to right looking downstream.
2. Digitise PO lines in channels and basins through the inundated area (main flow area) only. Create separate lines for the other inundation areas across the floodplain.

3. PO lines should be aligned roughly perpendicular to the direction of flow.
4. Lines to be added at the following locations, as a minimum
    a) immediate upstream and downstream of key cross drainage structures;
    b) through structures (digitise a QA line through the structure from bank to bank); and
    c) at areas of major interest (for example, at detention basins etc.).
5. Attributes to be used are H (water level), Q (flow), QA (flow area) and V (velocity).

PO Points

6. Add points at 50 meters intervals along channels and Overland flowpaths.
7. Add point at immediate upstream and downstream of dams and major structures.
8. At other key locations outside of the areas mentioned above.
9. Attributes to be used are H (water level) and V (velocity).

6.4.7 Model Validation

The pre-developed case (base case) model results must be validated against those from Council adopted flood models. To be specific, the 1% AEP peak flows and flood depths estimated by the applicant's pre-development flood model are not significantly different from the values derived from Council's flood models.

Model verification and establishing the base case model must be completed to satisfy the following minimum requirements:

For the base case model to be acceptable to Council, the modelled pre-development peak flows, and flood depths (at nominated locations within and in the vicinity of the subject site) should not differ from values extracted from Council's flood models by ±20%.

Council can provide the relevant 1% AEP peak flows and flood depths to the applicant but is subject to standard Council fees and charges.

6.4.8 TUFLOW Modelling Quality Assurance Log

The applicant to use a modelling log to maintain a record of model development, traceability, and quality assurance. An example of standard modelling quality assurance log is provided in Appendix – A.

The modelling log is to be submitted with each draft modelling review and the final model.

6.4.9 TUFLOW Modelling Checklist for Submission

Council's standard TUFLOW submission checklist template is added in Appendix C. Applicants to complete the checklist by adding items as required and attached a copy of the submission checklist with each submission.

- Details of the event modelled (duration, events, etc.); and
- Changes since last simulation.

The modelling log is to be submitted with each draft modelling review and the final model.

6.5 XP-RAFTS

The minimum requirements for XP-RAFTS modelling are as follows:

- The latest version of the software is to be used;
- Model simulations are to be run for the 50%, 20%, 10%, 5%, and 1% AEP storm, and the Probable Maximum Precipitation (PMP) events for study area;
- Review and confirmation of the impervious percentage used in the original RAFTS models are required;
- Use consistent naming for nodes and links;
- Apply Council's blockage policy for outlet structures, that is, AR&R 2019;
- Use AR&R 1987 rainfall;
- Calibration of the model to actual rainfall events and flows may be difficult owing to the lack of accurate flow data, flood records and appropriate comparison to historic land use and is therefore not required as part of this project. However, in the absence of calibration and verification data, model parameters must accurately reflect the existing physical conditions of the study area;
- Sensitivity analyses shall be carried out to assess how much influence the model parameter values have on the results. The principal parameters are rainfall losses, catchment storage, lag, and roughness;
- Ensure that the developed model will result in no worsening of flows;
- Models shall be run for a range of storm durations sufficient to identify the critical duration;
- Utilise the developed model for the sizing and configuration of the proposed basins and the culvert crossings, and other outlet structures; and
- Provide details of the assumptions on which the model is based.

6.5.1 RAFTS Modelling Quality Assurance Log

The Applicant is to use a modelling log to maintain a record of model development, traceability, and quality assurance. The format of the modelling log is up to the applicant. However, a spreadsheet should be developed for this purpose. Typical details to be entered into the log are:

- Names of RAFTS files;
- Date of simulation;
- Details of the event modelled (duration, events, etc.); and
- Changes since last simulation.

The modelling log is to be submitted with each draft modelling review and the final model.

7. Modelling Data Available with Council

Council currently possesses hydrologic and hydraulic models covering catchments, and subcatchment within the local Government area. If requested, Council can provide copies (or extracts) of these models in accordance with the following.

a. Council policies on the provision of flood information/models; and

b. Council's adopted Fees and Charges Schedule.

The available models or extracts (relevant information) can be used by the applicant or their consultants for the purpose of developing site-specific models. in doing so the applicant has to review and amend (as required) these models to develop a pre-developed (base case) model for their respective development site.

8. Assessment and Approval Process for H&H Modelling

[Section to be elaborated soon]

Applicants are requested to follow the necessary steps outlined below to ensure a smooth approval process for their flood modelling submission.

Step 1. Pre-lodgement Meeting

Applicants and their consultants are encouraged to seek Council's advice regarding modelling guidelines, requirements (e.g., software usage), data availability, etc., before initiating any modelling works related to a Development Application (DA) or other design tasks. This can be accomplished through a pre-lodgement meeting with representatives from Council's Development and Waterways teams.

Step 2. Acquiring Council Data

Flood models, flood modelling data, and Geographic Information System (GIS) data (available to Council) can be obtained by adhering to Council's standard fees and charges.

Step 3. Establishing the Pre-developed (Base) Case Model

Applicants should develop a pre-developed case model that complies with Council standards and the provided guidelines before submitting it for Council's review. Refer to Section 6.4.7 for additional information on validating the pre-developed case model. The applicant/consultant should not proceed with post-development flood modelling until Council approves the pre-development (base case) flood model.

Step.4. Lodgement

The developed base case model, along with a comprehensive modelling package, including civil design plans, architectural drawings, and a copy of the modelling report, all representing the same proposal, must be submitted to Council for review. Council will initiate the review process only when all the necessary information is received, ensuring it meets the submission requirements and standards.

Step 5. Assessment by Council

Upon receiving the full modelling package and associated information, Council will initiate a review of the submission. Applicants should allow a minimum of three weeks for small-scale development and four weeks for large-scale development for Council's review.

Step 6. Revisit Pre-developed Case Model

Applicants are required to address Council's comments and update the pre-development model accordingly. All significant comments related to the pre-development (base model) model must be resolved before conducting a flood impact assessment of the proposed development.

Step 7. Council Approving the Pre-developed Case Model

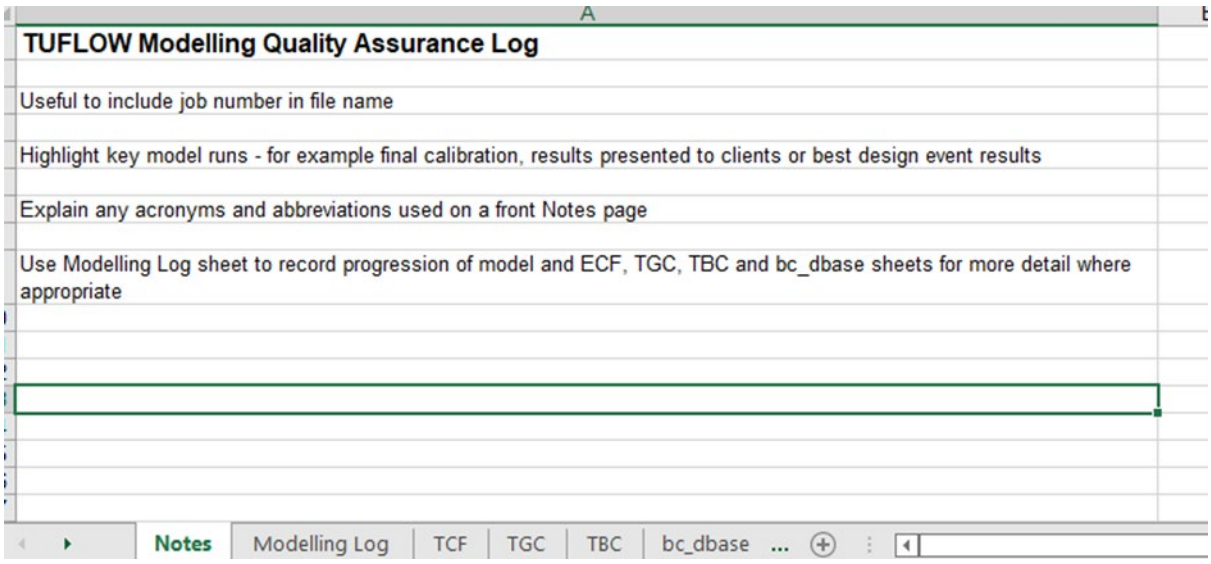
Council will grant approval to the Pre-developed Case Model once the final version of the package, meeting Council's requirements, is received. Applicants can then proceed with the post-development modelling and flood impact assessment (FIA).

Step 8. Establishing the Post-developed Case Model and Impact Assessment

Repeat Steps 3 to 6 until Council is reasonably satisfied with the outcome and endorses the submission.

Appendix A. Example TUFLOW Modelling Quality Assurance Log

Copy of the example spreadsheet can be obtained from Council. attributes of the example spreadsheet are second show screenshot below.



The spreadsheet to be used to include the following Details:

- Useful to include job number in file name
- Highlight key model runs - for example final calibration, results presented to clients or best design event results
- Explain any acronyms and abbreviations used on a front Notes page
- Use Modelling Log sheet to record progression of model and ECF, TGC, TBC and bc_dbase sheets for more detail where appropriate

Appendix B. Modelling Buildings in TUFLOW_Details of the Preferred Option.

Each building located within or adjacent to an overland flow path should be represented in the TUFLOW model using a completely impervious “wall” along the upstream face of each building. This approach is shown conceptually in Plate 1. An example of the building wall representation is provided in Plate 2. As shown in Plates 1 and 2, the downstream wall of each building was left open to allow rain falling on each building to “runoff” and contribute to catchment runoff.

Plate 2 also provides an example of overland flow velocity vectors in the vicinity of the buildings (the velocity vectors show the direction and speed of the overland flows). The velocity vectors show that the upstream walls provide an appropriate representation of the redirection of flow around buildings. It also provides a reliable representation of the increase in flow velocities as water is “squeezed” between buildings, thereby helping to ensure the flood hazard is suitably quantified.

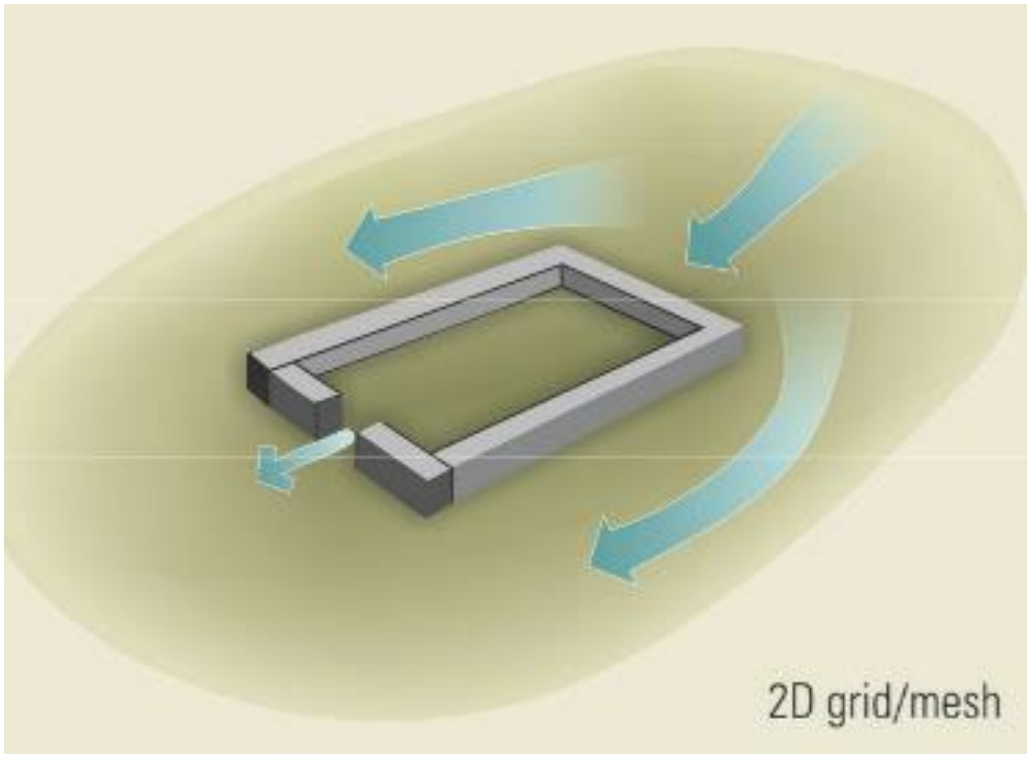


Plate 1. Conceptual representation of buildings in TUFLOW model, using the upstream building walls to deflect the flow and an “open” downstream wall to allow water to be stored in the building and to allow direct rainfall to “escape” (Engineers Australia, 2012)

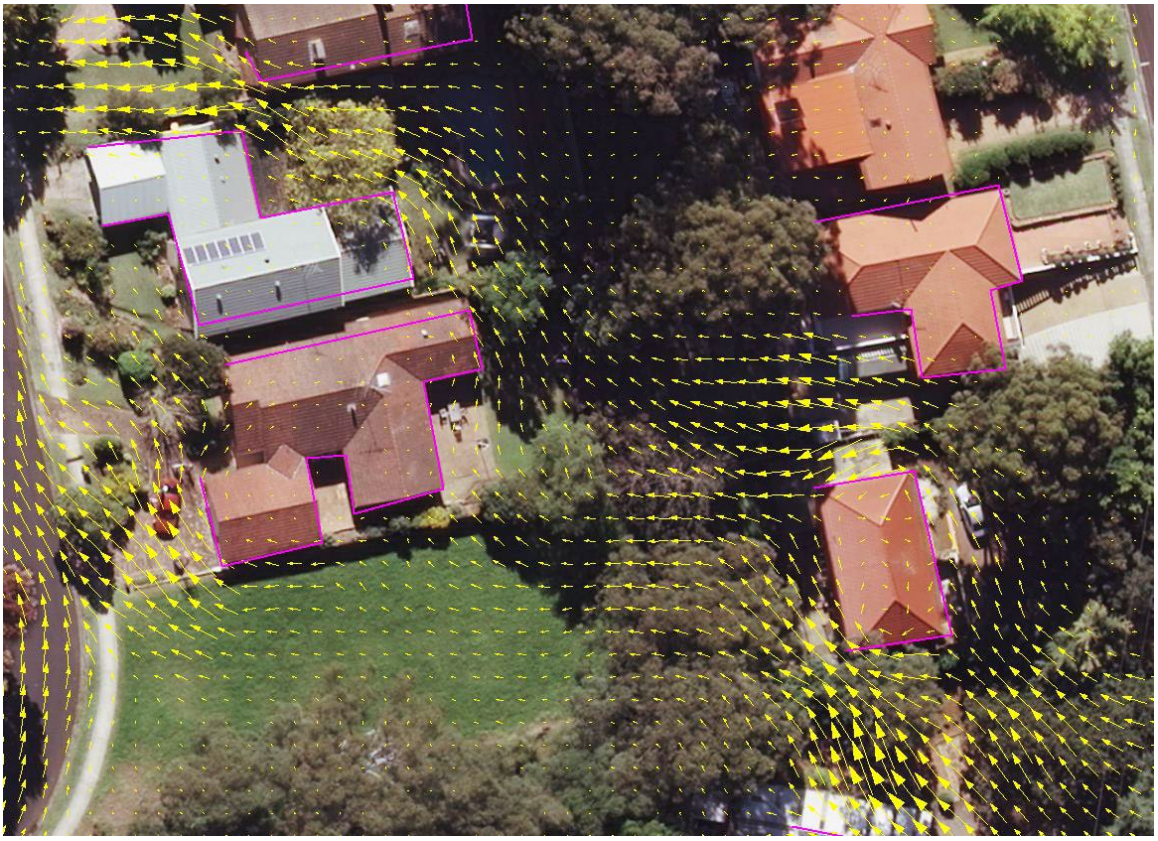


Plate 2 Example of upstream building wall representation (purple lines). Velocity vector arrows (yellow) show the walls force the water to move around and between buildings.

Appendix C. Checklists for Submission - TUFLOW Modelling

Table C1. TUFLOW Modelling Checklist

Item Number	Submitted	Meeting	Approved	TUFLOW Submission Checklist (Items to be added as required)	Comments
P0	□	□	□	Readme files identifying; • Purpose of submission and accompanying DA number; • Changes since previous version; and • Folder structure.	
P1	□	□	□	Model control files *(.tcf, .tgc, .tbc, .tef*) simplified with scenario for assessment. Files to include a comment header describing generally; • The purpose of the run; • The version the model has been based on; and • Changes made since last iteration of this model. All commands to have a comment describing their purpose.	
P2	□	□	□	Model input files provided for critical event and duration using “copyall” function. This produces a **.zip* file with all GIS files and DEMs called in running the model. *To be confirmed post preliminary check for completeness.*	
P3	□	□	□	Modelling log *(*.xls*) replicating comments in control files. Refer to Appendix-A for further detail.	
P4	□	□	□	Figures showing model setup and preliminary results. The figures are to generally include; 1. Manning’s Roughness; 2. Model Topography; 3. Hydraulic Structures; 4. PO Points and Lines;	

				5. New 1D Pit and Pipe Networks (if applicable); 6. Flood Depth and Elevation for Critical Event(s);	
P5	□	□	□	GIS layers to include; • Figure workspaces used for Item P4; • Inputs used in creating the TUFLOW scenarios; and • Filtered results grids.	
P6	□	□	□	Check files for critical event and duration runs.	
P7	□	□	□	Results files for critical event and duration runs.	
P8	□	□	□	Civil drawings used in support of the submission.	
P9	□	□	□	Updated correspondence and reports (as required).	
P10	□	□	□	Event and duration list (Table – 2)	

Table C2. Events and Durations Checklist

AEP / min	25	30	45	60	90	120	180	270
39.35%								
18.13%								
10%								
5%								
1%								
PMF								

Appendix D. Standard TUFLOW Materials File (.tmf)

Materials_Sample_THSC.tmf

```
! Comments and blank lines are allowed in this file
! First value is the Mat value
! Second is the Manning's n value
! Maximum of 20 different materials
!ID, n, IL, CL, y1, n1, y2, n2, 10, 2, f, IL, CL, Reasoning ,

1, 0.035, 15.0, 2.5 ! floodplain high grass
2, 0.015, 1.0, 0.0 ! roads
3, 0.04, 15.0, 2.5  ! light vegetation
4, 0.05, 15.0, 2.5  ! medium vegetation
5, 0.10, 15.0, 2.5  ! dense vegetation
6, 0.025, 0.0, 0.0  ! dams with little or no vegetation
7, 0.3, 0.0, 0.0   ! buildings or significant houses within the floodplain
8, 0.025, 0.0, 0.0   ! buildings or significant houses outside the floodplain
9, 0.04, 15.0, 2.5  ! waterway with minimal vegetation
10, 0.06, 15.0, 2.5 ! waterway with medium vegetation, raingarden
11, 0.08, 15.0, 2.5 ! waterway with dense vegetation
12, 0.30, 1.0, 0.0 ! Buildings, gardens, fences combined (as per ARR Project 15 Table 10-3) (Range 0.08-0.3)
13, 0.025, 5.0, 1.0 ! open space, sports fields, etc
14, 0.07, 15.0, 2.5 ! drainage channel
15, 0.15, 15.0, 2.5 ! very dense vegetation for sensitivity analysis
16, 0.04, 15.0, 2.5 ! rock armoring at culvert outlet
17, 0.02, 15.0, 2.5 ! bare construction site
18, 0.012, 1.0, 0.0 ! concrete surface
```

Appendix E. Example DRAINS Modelling Quality Assurance Log

DRAINS Modelling Log

Project Title:	
Modellers:	
Developed from earlier models:	
DRAINS version:	
Location of model files:	
Location of results:	

Run ID (Optional)	Date	File Name	Changes	Results/Comments
			Use this column to note the changes since previous version including file names where appropriate	Comments on run and results